

Mohamed Hussein

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Principal/Senior FullStack Software Engineer/Architect

Frisco, Texas, United States

Amazon AWS Solution Architect, Microsoft Azure Architect, and OutSystems Certified Full Stack .NET Developer with 18+ years hands-on experience across Telecommunications (OSS/BSS), Healthcare, Government, Real Estate, and Financial. Expert in .NET, Python, SQL, and API design, with proven cloud expertise in both Azure and AWS. Demonstrated expertise in building scalable, microservices-based, and event-driven solutions such as government e-Services, patient self-management apps (HL7), and ISP subscriber management solutions leveraging TMForum frameworks. Proficient in both front-end and back-end development including Angular, VueJs, .NET, ASP.NET, MVC, OutSystems, SharePoint, as well as infrastructure troubleshooting and data engineering processes automation (ETL and related workflows). Experienced in Azure and AWS cloud platforms. Certified in Big data and machine learning.

Skills

- C# | .NET | .NET Core | [ASP.NET](#) | Entity Framework (EF) | Python | Java | Node.js | JavaScript | C++ | C | MSSQL | MySQL | VueJS | Angular | jQuery | TypeScript | React | Git | OutSystems | MS SharePoint | BizTalk | HTML | CSS | PHP | Mobile | Cordova | Android | iOS | OpenText Documentum | Telecom broadband OSS/BSS | Powershell | Azure Web Apps | WPF | XAML | MVC | VS | Copilot | Cursor
- Web API | REST | SOAP | WCF | Service-oriented Architecture (SOA) | FastAPI | Security | OWASP | Oracle Cerner | HL7 | FHIR | DICOM | HIPAA
- AWS | Azure | GCP | Cloud Computing | CI/CD | DevOps | Unit Testing | TDD | OOP | AWS CloudFormation | Terraform | Bicep | Docker | Kubernetes (k8s) | Automatic Deployments | Kafka | ETL | SSIS | SSRS | Active Directory (Azure Entra ID) | VMWare | Azure Application Insights | AWS CloudWatch | Snowflake | Hadoop | Spark | Hive | Cloudera | AI/ML | OData (Open Data Protocol) | DMF (Data Management Framework) | Business Events)
- Microservices | Distributed Systems | Frontend | Backend | Full-Stack | Data Engineer | English | Arabic
- HTTP | HTTPS | FTP | SFTP | TCP/IP | JDBC | SSL | OAuth, Encryption | DNS | POP3 | IMAP | API Gateway | API Management | Load Balancer | WAF
- People Management | Supervision | Team management | Leadership | Agile | Scrum

Academic Projects

Agentic AI & RAG Pipeline

[Maharishi International University](#)

Cairo, Egypt 04/2025 - 05/2025

Designed and built an end-to-end retrieval-augmented generation system (RAG) using AWS S3 for storage, AWS Lambda for orchestration, Amazon EC2 for model inference, and a managed vector database for semantic search. Leveraged local LLM fine-tuning with Hugging Face Transformers, and delivered results via RESTful APIs and front end—cutting average query latency by 70 %.

Technologies used: Python, AWS S3, AWS CloudFront, AWS Lambda, Amazon EC2, Vector database, Hugging Face, Git, RESTful APIs

Stock Analysis

[Maharishi International University](#)

Cairo, Egypt 09/2025 - 11/2025

Architected and developed a production-grade stock market analysis platform featuring an end-to-end Retrieval-Augmented Generation (RAG) pipeline. Implemented AWS S3 for storage, AWS Lambda for orchestration, Amazon EC2 for model inference and API hosting, and a managed vector database for semantic search and financial insight retrieval. Leveraged Hugging Face Transformers for fine-tuning local LLMs, integrated with .NET Core backend services and a FastAPI inference layer. Exposed intelligent insights through secure RESTful APIs and a responsive front-end interface.

Technologies: .NET Core, Python, Scala, AWS S3, AWS CloudFront, AWS Lambda, Amazon EC2, Redis, Vector Database, Hugging Face Transformers, FastAPI, API Gateway, Kubernetes, Docker, Kafka, AWS Secrets Manager, Git, CI/CD.

Experience

Principal Software Engineer
/Senior Engineering Manager

[Link Development](#)
[Client:Federal entities](#)

Cairo, Egypt 01/2019 - 04/2024

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- Implemented web application best practices in Angular, Javascript, HTML, Bootstrap, CSS front-ends and .NET Core, Web APIs, LINQ, IIS, REST, SOAP, OAuth2, Identity Server, Git with SQL Server, safeguarding against XSS, CSRF, SQL Injection, and authentication/authorization vulnerabilities. Led Dashboard and reports using Power BI.
- Leveraged .NET Core and Entity Framework Core with SQL Server to streamline database schema design, optimize queries, and improve data access performance across multiple applications.
- Integrated test frameworks to implement unit, integration, and automated testing, ensuring high software quality and maintainability.
- Led the design and development of enterprise-grade applications for various government entities, generating millions in annual revenue.
- Designed and implemented scalable systems and background workers to overcome legacy system limitations, leveraging OutSystems Traditional, Reactive, OutSystems Mobile, and Identity Server to serve a large population of patients, residents, and citizens.
- Directed end-to-end project delivery—overseeing requirements, building Angular front-ends and .NET back-ends, executing Azure DevOps CI/CD pipelines, DevSecOps, OWASP, and managing production deployments and post-go-live support.
- Enhanced system capabilities by integrating with external systems (Payment Gateways, Identity Server, eSeal) via REST and SOAP, streamlining authentication, payment processing, and document management through OpenText Documentum.
- Leveraged Azure DevOps to drive agile/SCRUM processes—managing Boards, user stories, backlog prioritization, sprint planning, task assignments—and generated burndown charts to track sprint progress, team velocity, and ensure transparent delivery.
- Engaged directly with customers to gather technical requirements, address challenges, and ensure satisfaction with delivered products and services.
- Provided technical leadership and mentorship to development teams, conducting pull-request reviews, load/stress testing, and coaching on best practices for code quality and performance.
- Collaborated with the pre-sales team on estimation, technical consulting, demos, and prototypes, contributing to the successful acquisition of tenders.
- Designed and implemented an enterprise EMR integration layer using .NET and HL7 v2.x messaging (ADT, ORM, ORU) over MLLP, and DICOM standards, enabling real-time, HIPAA-compliant patient data exchange with Cerner and other Medical systems. Applied PCI compliant for payments for self patient management portal.
- Implemented a cloud-native government portal using Azure Function Apps and Azure API Management, integrated with Azure SQL Database, Azure Data Factory, Azure Service Bus, Azure Key Vault, Azure Application Insights, Azure Load Balancer, Azure Web Application Firewall (WAF), Azure Virtual Machines, Azure Content Delivery Network (CDN), and Azure App Service, ensuring secure API endpoints via OAuth2 and managed identity.
- Led and participated in a large-scale migration project from legacy systems using ETL pipelines, ensuring accurate data transformation, cleansing, and seamless integration between old and new platforms.
- Applied SOLID principles and design patterns (Factory, Singleton, Repository,...) to architect maintainable enterprise applications.
- Architected and delivered cloud-native enterprise solutions using microservices and event-driven patterns, ensuring scalability, security, and high availability.
- Architected and implemented the Azure Integration Services platform leveraging Entra, Logic Apps, API Management, Service Bus, Azure Functions, and Key Vault (KV), event-driven integrations with Dynamics 365, D365 Finance & Operations, SAP, Salesforce and other enterprise systems using Open Data Protocol(OData), Data Management Framework(DMF), Business Events. Implemented Zero-Trust controls, Managed Identities, Private Endpoints for full network isolation, NAT Gateway and Firewalls.
- Defined reference architecture and reusable integration patterns with enforced API Management policies (OAuth2/JWT validation, throttling, retries, schema validation). Built observability dashboards using Kusto Query Language (KQL) and Application Insights for failure tracking, Dead Letter Queue (DLQ) monitoring, and latency analysis. Designed iPaaS integrations on different platforms (ex: Boomi).

Staff Software Engineer / Development Manager

[Link Development](#)
Client:Federal entities

Cairo, Egypt 01/2016 - 12/2018

- Led the design and development of multiple enterprise-level applications providing e-Services driving millions of revenue every year using the technologies of MS SharePoint, Angular, C#, .NET, MS SQL Server, Integration with MS CRM .
- Designed and implemented intranet and background workers for employee to communicate with each other
- Implemented Performance and caching handling through Azure CDN and Redis. Also, using Azure Key Vault (KV) to secure important keys and secrets other Azure cloud technologies. Utilized Azure Functions, API Gateway in implementing core functions and API interface, Integrated apps authentications using Entra ID (Active Directory) and OAuth2
- Continuous Integration/Deployment Pipeline Integration, pull requests, code reviews, load/stress testing, unit/integration testing using Azure DevOps

Technical Lead

[Link Development](#)
Client:Telecom Operator

Cairo, Egypt 01/2012 - 12/2015

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- Led the design and development of multiple enterprise-level applications providing multiple services and features to DSL (Digital subscriber line) customers driving millions of revenue every year using the technologies of C#, .NET, MS SQL Server, Integration with MS CRM, Integration with MS Great plains (now called MS Dynamics 365 Business Central).
- Designed and implemented enterprise telecom applications and background workers to fulfill customers requests.
- Participated in implementation of Self management portal for landline customers that leads to facilitating customer's requests easily.

Senior Solution Developer

[Link Development](#)
Client:Telecom Operator

Cairo, Egypt 01/2009 - 12/2011

- Designed and implemented enterprise telecom applications for one of the largest DSL, driving yearly revenue of millions of \$ using C#.NET, ASP.NET, MVC, Windows Communication Foundation (WCF), MSSQL, Javascript and jQuery.
- Automated and optimized business logic for the core systems for order capture and order fulfillment for telecom applications
- Participated in automated provisioning modules that helped telecom applications to work efficiently using .net LDAP, DSLAM protocols/commands providing instant provision to the end users.

Solution Developer

[LinkdotNET](#)

Cairo, Egypt 11/2006 - 12/2008

- Automated and optimized business logic for the core systems for order capture and order fulfillment for telecom applications using C#.NET, ASP.NET, MSSQL, BizTalk and Javascript..

Software Developer

[YAT Software](#)

Cairo, Egypt 08/2006 - 11/2006

- **Warehouse application:** Developed warehouse application to manage stock of free packages (ASP.NET, C#, MS SQL Server)

Education

Master of Science in Computer Science

[Maharishi International University](#)

Fairfield, Iowa, US 04/2024 - Present

Diploma in Data Science and Big data

[Nile University](#)

Giza, EG 09/2015 - 05/2016

Bachelor of Science from Faculty of Computers and Information

[Helwan University](#)

Helwan, Cairo,EG 09/2002 - 05/2006

- Major in Computer Science

Certificates

- **Solution Frameworks Overview:** Foundation Certificate for business architecture provided from TM Forum (2009)
- **Information Framework (SID) certification:** A Certificate for Common Information Model provided from TM Forum (2010)
- **Business Process Framework (eTOM) certification:** A Certificate for business-led approach to rationalize the processes that run your business provided from TM Forum (2010)
- **AZ-303: Microsoft Azure Architect Technologies:** Azure architecture certificate provided from Microsoft (2022)
- **Associate Reactive Developer:** Certificate provided from OutSystems (2022)
- **AWS Certified Solutions Architect - Associate:** Certificate provided from Amazon AWS(2024)

Training

- **TOGAF Foundation:** provided by SECC (2018)
- **TOGAF Level 2:** provided by SECC (2018)

Others

- **Award:** Won Employee appreciation